Harnessing GAN to Create Realistic FBS(False Base Station) Attack Data in 5G

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***Abstract*— The threat of false base stations (FBS) has evolved with the advancement of mobile communication, posing severe risks in the 5G era. FBSs can mimic legitimate stations, leading to various malicious activities like man-in-the-middle attacks, denial-of-service attacks, and spreading fake disaster messages. Detecting FBSs is crucial, but existing methods relying on machine learning face challenges due to insufficient datasets. To address this, Generative Adversarial Networks (GANs) are explored to augment limited datasets by generating realistic data. This paper investigates using GANs to enhance datasets for FBS detection, compares the similarity between generated and real datasets, and assesses detection accuracy. Furthermore, we trained machine learning models on GAN-generated data and utilized it to implement an FBS detection Network Function (NF) for 5G environments. Ultimately, we generated a dataset with a high similarity, with a Pearson correlation coefficient of 0.85, using GANs. We confirmed the possibility of utilizing this dataset for FBS detection and applied it in a 5G environment.**

Keywords—FBS, GAN, 5G, NF

I. Introduction

5G is the fifth generation of wireless cellular technology, which supports ultra-connectivity, ultra-high speed, and ultra-low latency. Considered a cornerstone of the Fourth Industrial Revolution, 5G is utilized across numerous industries ranging from healthcare, insurance, and construction to factories and agriculture. According to reports from Omdia, the global 5G subscriber count surpassed 1 billion by the end of 2022 [1]. As rapid commercialization is underway, 5G security is becoming increasingly critical.

With the advancement of mobile communication, the threat of false base stations (FBS) has also evolved. Such attacks have become even more severe as we enter the era of 5G. FBSs disguise themselves as legitimate ones, emitting strong signals to lure User Equipment (UE) and causing social disruption through activities like man-in-the-middle attacks, denial-of-service attacks, stealing personal information, spreading fake disaster messages, and malware attacks. In reality, the Bank of Thailand suffered losses of approximately 200 million THB in May 2023 alone due to malware messages propagated through FBSs [2]. In Korea, the absence of a disaster message authentication protocol leaves room for attackers to generate and disseminate fake disaster messages. Thus, FBSs pose threats to mobile subscribers through various attacks. Detecting FBSs is a crucial research area in the 5G era to counteract their attacks, necessitating the development of FBS detection systems.

Existing research on FBS detection implements machine learning, utilizing the Measurement Reports continuously generated by UEs as a dataset. However, FBS detection methods based on machine learning are only possible if the Measurement Report, i.e., the dataset, is sufficient. Numerous exceptional situations can occur in real life, making it difficult to secure an adequate dataset. Therefore, strategies are needed to detect FBSs even with limited datasets.

Generative Adversarial Networks (GAN) are generative deep learning models capable of generating data similar to actual data [3]. The adversarial training scheme is rapidly gaining attention in academia and industry for its ability to counteract domain shifts. Additionally, it is highly effective in generating new image samples, making it a powerful tool for enhancing image classification models.

Gathering an adequate number of images for a particular application remains difficult due to constraints such as subject type, image quality, geographical location, time period, privacy, and copyright status. [4]. This curbs the increasing use of generative models in fields such as medicine [5]. A significant reduction in the amount of data required. Therefore, it can potentially help many applications, such as mobile communication.

By introducing GAN, it is possible to generate data similar to actual data from limited datasets and detect FBSs based on machine learning. This overcomes the limitations of insufficient datasets in a 5G environment.

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Therefore, this paper experimentally investigated a method to use GAN to augment the limited amount of Measurement Reports, enabling the detection of the presence of FBS. The similarity between the generated Measurement Reports by GAN and the actual Measurement Reports was calculated and compared to determine how closely they resemble each other. Additionally, the detection accuracy using only actual Measurement Reports was compared with that using Measurement Reports generated by GAN to assess the significance of this approach. Additionally, leveraging the dataset generated by GAN, various machine learning models were trained to implement a Network Function (NF) capable of detecting FBS in a 5G environment.

II. Related work

When exploring existing research in the field, various methods for detecting Measurement Reports are available [6, 7]. However, there has yet to be an attempt to overcome limited datasets and attempt FBS detection using GAN. Nevertheless, similar attempts can be found in other fields. Desai, Shrinivas D. et al. [8] utilized GAN in the medical field, where a reliable labeled dataset is scarce. They generated breast cancer images using DCGAN and analyzed the similarity between the original and generated images, achieving a dataset with 87% similarity. S. Aissa and K. M. Tarek [9] generated datasets for electricity consumption prediction using Time GAN. Predicting electricity consumption is essential for effectively managing the power grid and ensuring a stable and reliable supply in the short and medium terms. However, real-world scenarios often face the challenge of limited datasets. To address this, Time GAN, which specializes in time series data, was used to generate data similar to the original. This approach led to a prediction model with 7.45 higher accuracy. Asghar, Usman, et al. [10] generated X-ray and CT image datasets to determine whether a patient has contracted COVID-19 using GAN. Previously, the dataset needed to be improved, leading to ineffective models due to overfitting. However, utilizing GAN to leverage more data, they created a model with 92% accuracy in distinguishing patients. Frid-Adar, Maayan, et al. [5] generated a CT image dataset of liver lesions using GAN. Adding synthetic data augmentation improved specificity from 88.4% to 92.4% and sensitivity from 78.6% to 85.7% compared to using only traditional data augmentation.

These previous studies have utilized GAN, especially in medical fields where reliable labeled datasets are scarce, to generate similar data and create high-accuracy models. Therefore, it is expected that GAN could be utilized to overcome situations with limited reliable datasets in FBS detection.

III. Background

A. Security Threat posed by False Base Stations

The major 5G security threat, FBS attacks, occurs worldwide. When summarized in a table, it looks like TABLE Ⅰ.

B. UERANSIM

We constructed a simulation environment using a 5G UE and RAN Open-Source called UERANSIM to obtain normal and attack datasets.

UERANSIM has been widely used in 5G simulation research [11, 12, 13]. However, the existing UERANSIM has limitations because it only partially adheres to 3GPP's official documents. Therefore, we upgraded UERANSIM by following TS 38.401 [14], which mandates the functional split of CU and DU in the base station and adding base stations that play the most crucial role of FBS in this study. UERANSIM has produced significant results in numerous studies, and by thoroughly upgrading it to adhere to official documents, it is reasonable to use it as a real-world scenario simulator.

TABLE I. SECURITY THREAT POSED BY FBS

Security Threat	Explanation
Passive Attack	The false base station detects nearby wireless signals and stores the parameters and messages of neighboring cells.
Active Attack	An attack where a false base station masquerades as a legitimate one, receiving handover mobility management procedure messages from users.
Downgrade Attack	An attack that downgrades the quality of a user's device by forcing it to use a previous-generation network.
Location tracking Attack	An attack that tracks users' precise location within the range of a false base station based on collected user information.
Insertion Attack	An attack is when an attacker inserts arbitrary messages into user messages.
Information leakage	An attack that exploits the characteristics of user devices to select the stronger signal causes users to connect to a false base station, which then collects their IMSI.
Spoofing Attack	UEs that follow the LTE standard trust the authenticity of messages received before EPS-AKA, so it is possible for an attacker to deceive the user by sending false information.
Relay Attack	The attacker uses a false base station and malicious user equipment (UE) to lure the victim UE into connecting. The malicious UE then relays the victim's messages to a remote legitimate base station, intercepting the communication.

C. Measurement Report

For our experiment dataset, we utilized Measurement Reports. As usual FBS detection based on machine learning typically operates, we generated the dataset centered around RSRP (Reference Signal Received Power) within the Measurement Reports [6]. Measurement Reports are generated by UE whenever specific conditions specified in 3GPP TS 38.331 are met [15]. They contain information about each base station's PCI (Physical Cell Identity) around the UE, along with RSRP, which denotes signal strength. FBSs typically transmit stronger signal strength than legitimate base stations to lure UE into connecting to them. Machine learning algorithms are trained on this data to recognize patterns and detect such anomalies.

D. Pearson Correlation Coefficient

After generating the dataset with GAN to compensate for the insufficient dataset, it is crucial to compare how similar the generated dataset is to the original dataset. If they are dissimilar, the generated dataset becomes meaningless. Therefore, we measured the similarity between the dataset generated by GAN and the original dataset using the Pearson correlation coefficient. The correlation coefficient is one of the most commonly used methods to validate similarity (correlation coefficient ($-1 \leq CC \leq 1$)). The closer the absolute value of the calculated coefficient is to 0, the lower the similarity, and the closer it is to 1, the higher the similarity.

The interpretation of the Pearson correlation coefficient's results can be shown in Table II.

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TABLE II. INTERPRETATION OF CORRELATION COEFFICIENT

Scale of Pearson correlation coefficient	Interpretation
$0.00 \le \lvert r \rvert \le 0.19$	Very low correlation
$0.20 \le \lvert r \rvert \le 0.39$	Low correlation
$0.40 \le \lvert r \rvert \le 0.59$	Moderate correlation
$0.60 \le \lvert r \rvert \le 0.79$	High correlation
$0.80 \le \lvert r \rvert \le 1.00$	Very high correlation

IV. EXPERIMENT

One prominent method for detecting FBSs involves using artificial intelligence models. These AI models identify anomalies within extensive RRC (Radio Resource Control) data to detect FBSs based on their learned patterns. However, sufficient training data is essential to effectively train and utilize artificial intelligence models. Nevertheless, FBS attacks are often sporadic and occur within short timeframes, posing limitations on collecting an adequate quantity of attack data in real 5G environments.

To overcome these limitations, generative artificial intelligence can be employed. Generative AI can learn from the limited amount of existing data and generate new data similar to the existing data in the desired quantity. By training with a small amount of FBS attack data using generative AI techniques such as GAN, it is possible to generate a large amount of similar attack data. Consequently, this approach can enhance the performance of FBS detection by providing a substantial amount of synthetic attack data.

A. *Simulation through UERANSIM*

We obtained the FBS attack dataset using the updated UERANSIM, which incorporates the new functionality we added. The simulation environment is as follows:

- 9 Base Stations in a 2250 x 2250 range
- Base Station No.9 is the False Base Station
- UE changes direction every 10 seconds
- UE moves at a speed of 5m/s – 10m/s

As shown in Fig. 1, 66,595 attack data were prepared through simulation. Subsequently, preprocessing was conducted to enable GAN to train on the data.

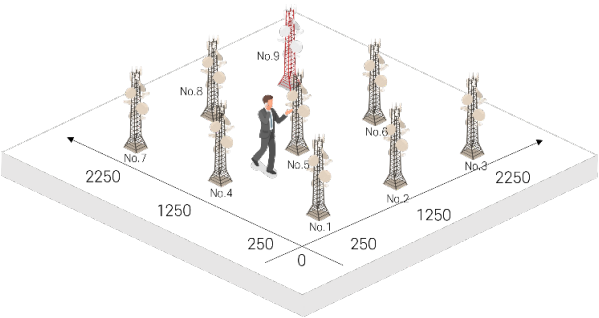
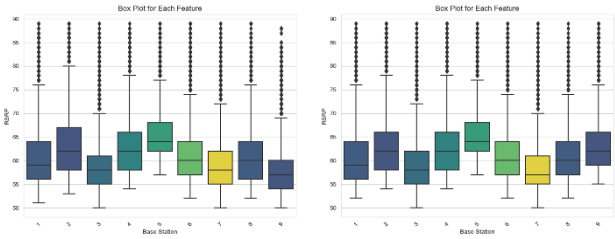


Fig. 1. Simulation environment.

Fig. 2 shows the variance of RSRP values when only Normal Base Stations are present (a) and False Base Stations are present (b). While the RSRP values of other base stations are similar, it can be observed that only the RSRP values of Base Station 9, set as a False Base Station, exhibit a difference in distribution between (a) and (b). This visually confirms that False Base Stations transmit stronger signal strength than other normal base stations to lure User Equipment (UE).



(a) Original Datasets (b) False Base Station Datasets

Fig. 2. Data Distribution.

B. *Training GAN*

The advantage of GAN is its ability to learn from a small amount of data. This is why we chose GAN over other generative methods in this paper. Therefore, we trained GAN to generate data using only a portion of the 66,595 attack data generated through simulation. However, since the term 'small amount of data' is subjective, it is necessary to determine how much data is needed to derive meaningful results. Therefore, we used Pearson correlation coefficients to compare the similarity between the original and training data while increasing the training data by 10,000 at a time.

TABLE III. SIMILARITY BY THE AMOUNT OF TRAINING DATA

Amount of data	Similarity (Correlation Coefficient)
10,000	0.8476576196655823
20,000	0.8486576196655814
30,000	0.8478829035770166
40,000	0.8464628470014133
50,000	0.8469690457034459
60,000	0.8467601375927342

Table III shows that the highest similarity was observed when there were 20,000 training data, and after that, it was observed that the similarity decreased even with more significant amounts of data. This is due to the overfitting issue occurring within the GAN.

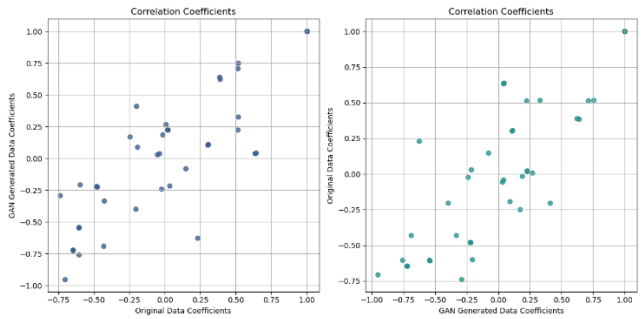


Fig. 3. Pearson Correlation Coefficients.

When training with the dataset of 20,000 samples that exhibited the highest similarity, the correlation between the generated dataset and the original dataset was approximately 0.85; according to TABLE II, this falls under Very high correlation. This indicates that the two datasets have a meaningful level of similarity.

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C. *Evaluating dataset efficacy by ML algorithms*

FBS detection is primarily conducted using machine learning algorithms. Training it on machine learning algorithms and validating its accuracy is essential to assess whether the generated GAN can be utilized for actual FBS detection. Thus, we used both the "original attack data generated through simulation" and the "dataset generated by injecting a small amount of data into the GAN" as attack data for FBS detection machine learning models. The normal data remained consistent and generated through simulation. Subsequently, we compared the accuracy of these two scenario simulations. Subsequently, we compared the accuracy of these two scenarios.

The total data used in the experiment was 40,000, comprising 20,000 attack data and 20,000 normal data.

TABLE IV. MACHINE LEARNING RESULT COMPARISON

MODEL	Original Data	GAN-Generated Data
CNN	94%	93%
SVM	81%	85%
XGBoost	95%	95%
Logistic Regression	75%	80%
KNN	80%	81%
Naïve Bayes	57%	64%
Random Forest	82%	81%

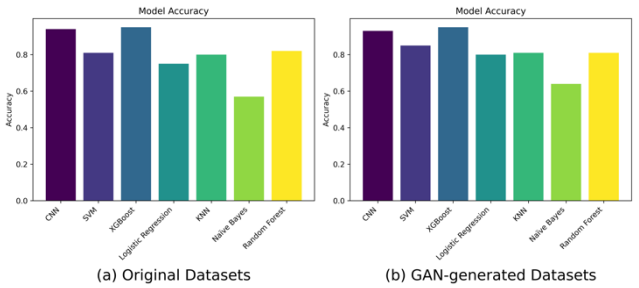

Fig. 4. Machine Learning Result Comparison

Ultimately, in all AI models, FBS data generated by GAN was detected at a level similar to the original attack data when compared.

V. IMPLEMENTATION OF FBS DETECTION NETWORK FUNCTION IN A 5G ENVIRONMENT

Through the experiments, it was confirmed that the data generated by GAN is quite similar to real data, and the artificial intelligence model trained with GAN data detects attack data with similar accuracy to the model trained with real data. Therefore, it is feasible to implement a Network Function (NF) using such artificial intelligence models to detect rogue base stations in real-time in a 5G environment.

Multiple models trained on NF were added to enable real-time detection. The models used include SVM, KNN, Decision Tree, Random Forest, XGBoost, and Naive Bayes. Each model was trained with normal data and attack data generated by GAN.

Fig.5. represents CU, Fig.6. represents nine base stations, Fig.7. UE, and Fig.8. represents the implemented FBS detection NF. The UE continuously sends Measurement Reports to the CU. The NF determines the presence of FBSs based on the Measurement Reports received from the CU and displays the results on the screen. In Fig.6., we can see that the NF outputs the presence of FBSs and their accuracy predicted by each AI model along with the Measurement Report.

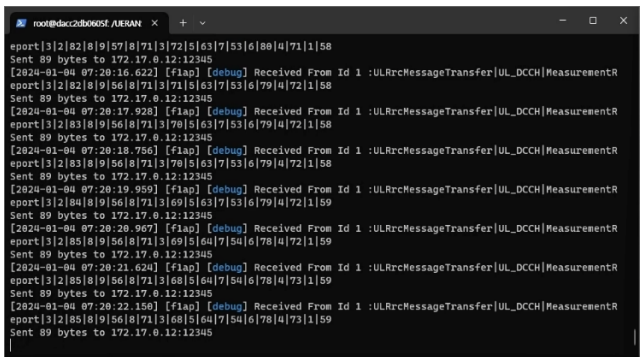

Fig. 5. gNB; CU.

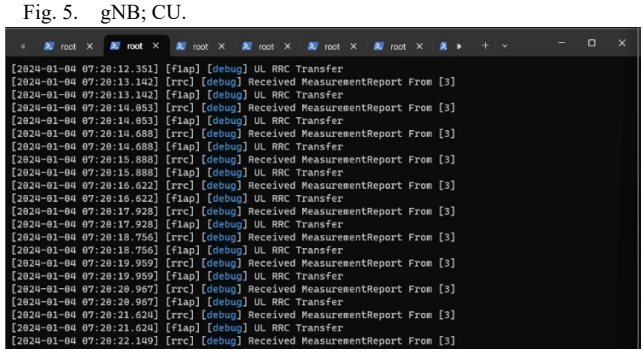

Fig. 6. gNB; DU.

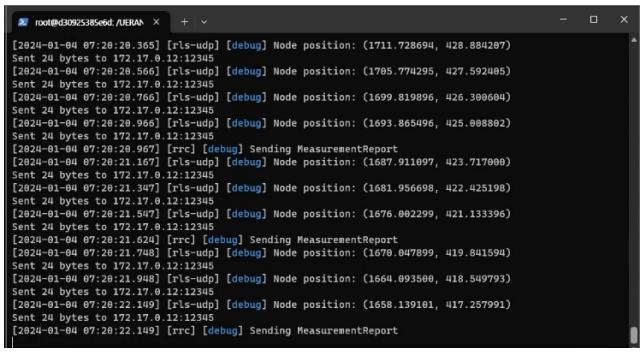

Fig. 7. User Equipment

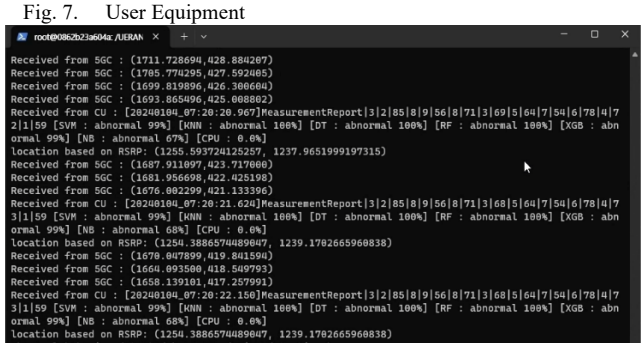

Fig. 8. 5G Core; Network Function

VI. CONCLUSION

This paper conducted various experiments on utilizing Generative Adversarial Networks (GANs) for generating

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and applying training data for False Base Station (FBS) detection in a 5G environment. Initially, data was extracted from various simulated false base station attack scenarios, and then this extracted data was augmented using GANs to increase its quantity. Finally, the similarity between GAN-generated data and real data was compared using the Pearson correlation coefficient, and the possibility of utilizing GAN-generated data for FBS detection in actual artificial intelligence models was examined. Approximately 0.85 of very high correlation and high potential were confirmed, and GAN-generated data were used to train artificial intelligence models and implement a real-time FBS detection NF (Network Function). The study confirmed the potential of GAN-generated data for training AI models for real-time FBS detection, demonstrating its utility in environments where obtaining sufficient real-world data is challenging. Despite the success in detecting the presence of false base stations, the study noted limitations in precisely identifying which base station is false and its location, highlighting the need for more complex time-series data. Future research will address these limitations and explore other generative AI types.

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REFERENCES

[1] Ombida, “Growth of 4G over 5G in 2022 raises concern for 5G evolution.”, https://omdia.tech.informa.com/pr/2023/feb/omdia-growth-of-4g-over-5g-in-2022-raises-concern-for-5g-evolution (accessed April 24, 2024).

[2] Bangkok Post, "A new threat on the banking horizon.” https://www.bangkokpost.com/business/general/2611359/a-new-threat-on-the-banking-horizon (accessed April. 24, 2022).

[3] Goodfellow, Ian, et al., "Generative adversarial nets." Advances in neural information processing systems 27, 2014.

[4] Karras, Tero, Miika Aittala, Janne Hellsten, Samuli Laine, Jaakko Lehtinen, and Timo Aila. "Training generative adversarial networks with limited data." *Advances in neural information processing systems* 33 (2020): 12104-12114.

[5] Frid-Adar, Maayan, Idit Diamant, Eyal Klang, Michal Amitai, Jacob Goldberger, and Hayit Greenspan. "GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification." *Neurocomputing* 321 (2018): 321-331.

[6] Prajwol Kumar Nakarmi, Jakob Sternby, and Ikram Ullah. “Applying Machine Learning on RSRP-based Features for False Base Station Detection.” In Proceedings of the 17th International Conference on Availability, Reliability and Security (ARES '22). Association for Computing Machinery, New York, NY, USA, 2022, Article 60, 1–7.

[7] J. Jin, C. Lian, and M. Xu, "Rogue Base Station Detection Using A Machine Learning Approach," 2019 28th Wireless and Optical Communications Conference (WOCC), Beijing, China, 2019, pp. 1-5,

[8] Desai, S.D., Giraddi, S., Verma, N., Gupta, P., & Ramya, S.T., “Breast Cancer Detection Using GAN for Limited Labeled Dataset.” *2020 12th International Conference on Computational Intelligence and Communication Networks (CICN)* (2020): 34-39.

[9] S. Aissa and K. M. Tarek., "Time Generative adversarial network for the generation of electricity load data," *2023 International Conference on Control, Automation and Diagnosis (ICCAD)*, Rome, Italy, 2023, pp. 1-5

[10] Asghar U, Arif M, Ejaz K, Vicoveanu D, Izdrui D, Geman O., “An Improved COVID-19 Detection using GAN-Based Data Augmentation and Novel QuNet-Based Classification.” *BioMed research international* vol. 2022 8925930. 26 Feb. 2022

[11] Sylla, Tidiane, et al. "Emu5gnet: An open-source emulator for 5g software-defined networks." *2022 18th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*. IEEE, 2022.

[12] Salazar, Zujany, et al. "5greplay: A 5g network traffic fuzzer-application to attack injection." *Proceedings of the 16th International Conference on Availability, Reliability and Security*. 2021.

[13] Li, Yin-Chi, et al. "A Runtime Anomaly Detector via Service Communication Proxy for 5G Mobile Networks." *IEEE INFOCOM 2023-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. IEEE, 2023.

[14] 3GPP, "Technical Specification Group Radio Access Network; NG-RAN; Architecture description.", https://www.3gpp.org/DynaReport/38401.htm (accessed April. 10, 2024).

[15] 3GPP, "Technical Specification Group Radio Access Network; NR; Radio Resource Control (RRC); Protocol specification.", https://www.3gpp.org/DynaReport/38331.htm (accessed April. 10, 2024)

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